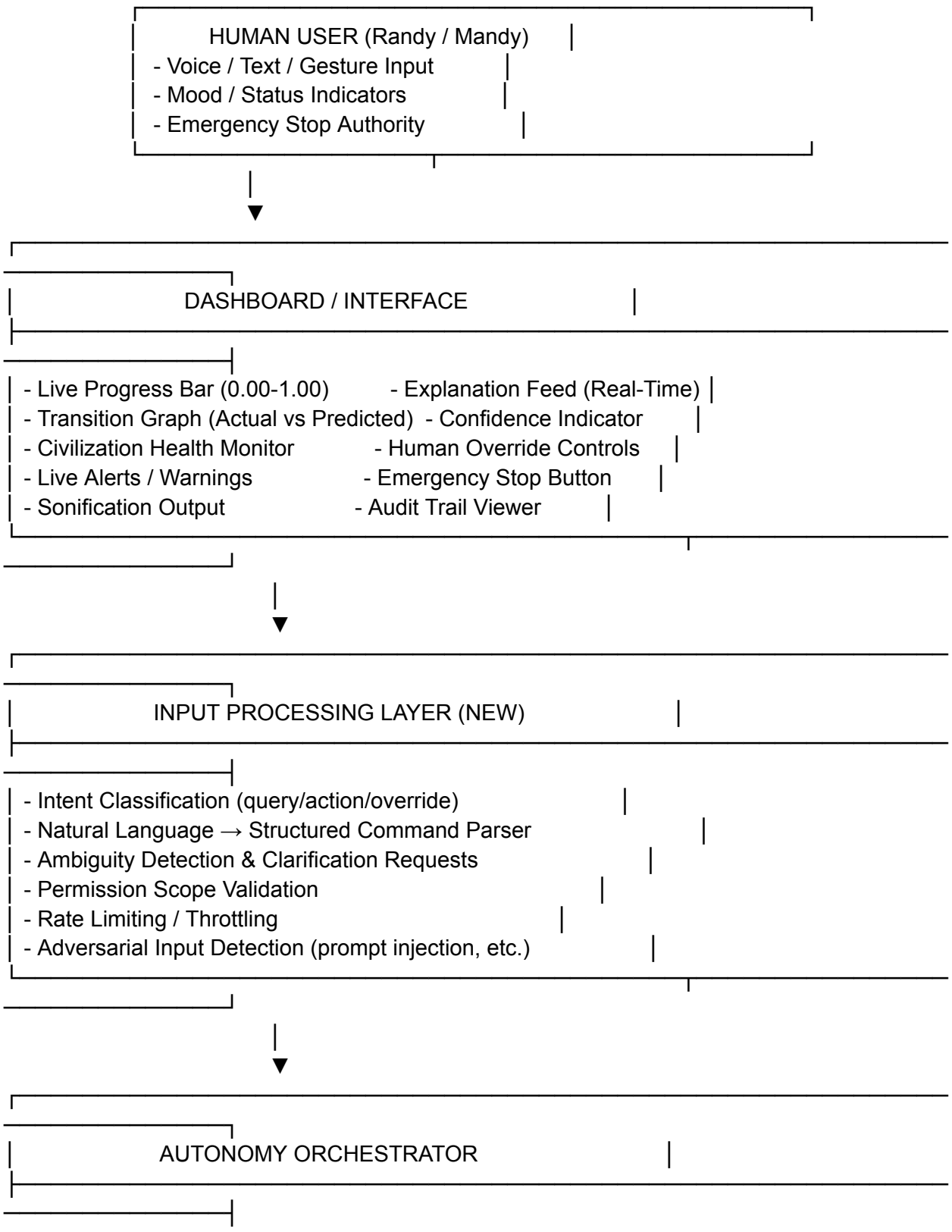
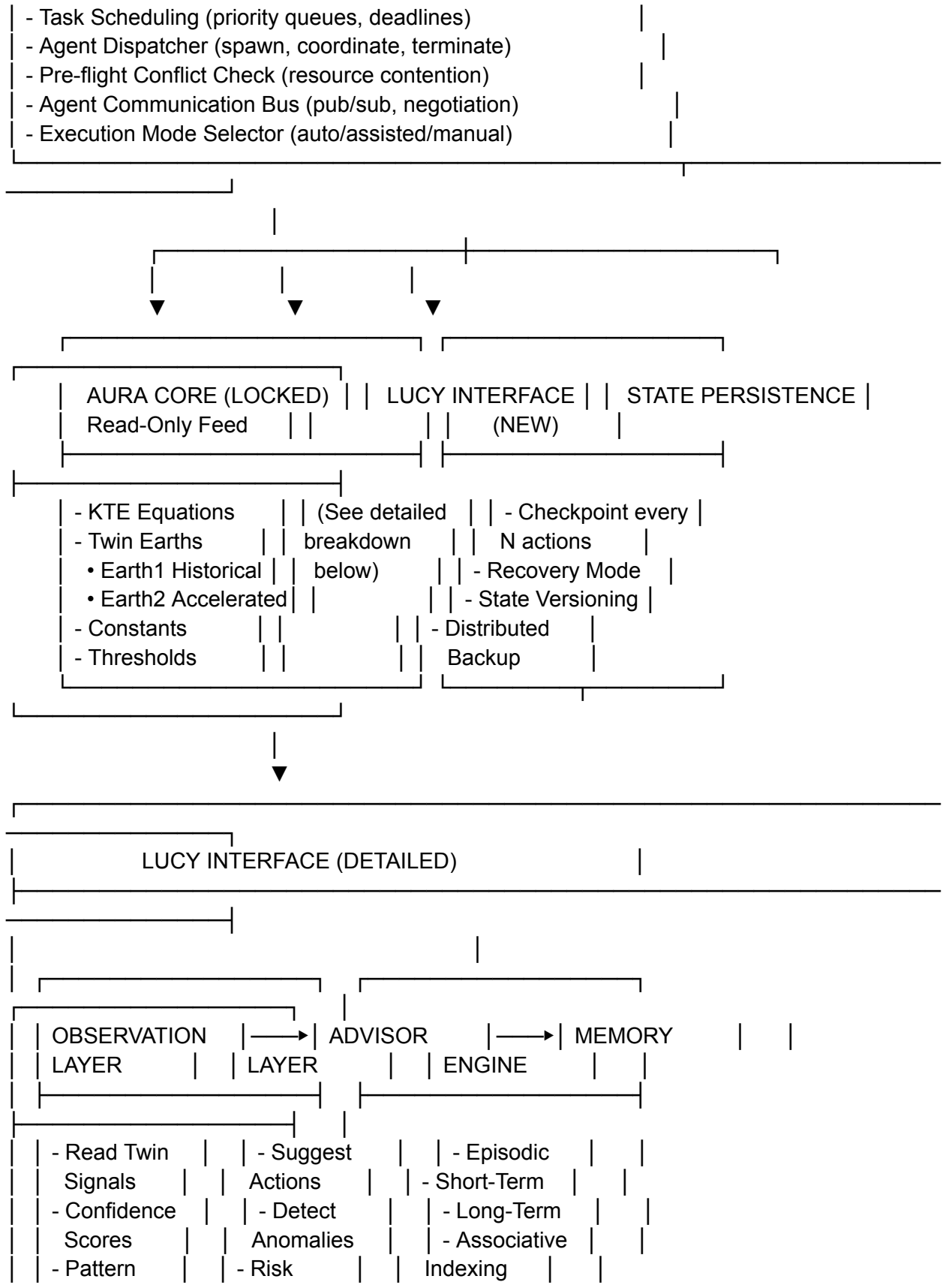
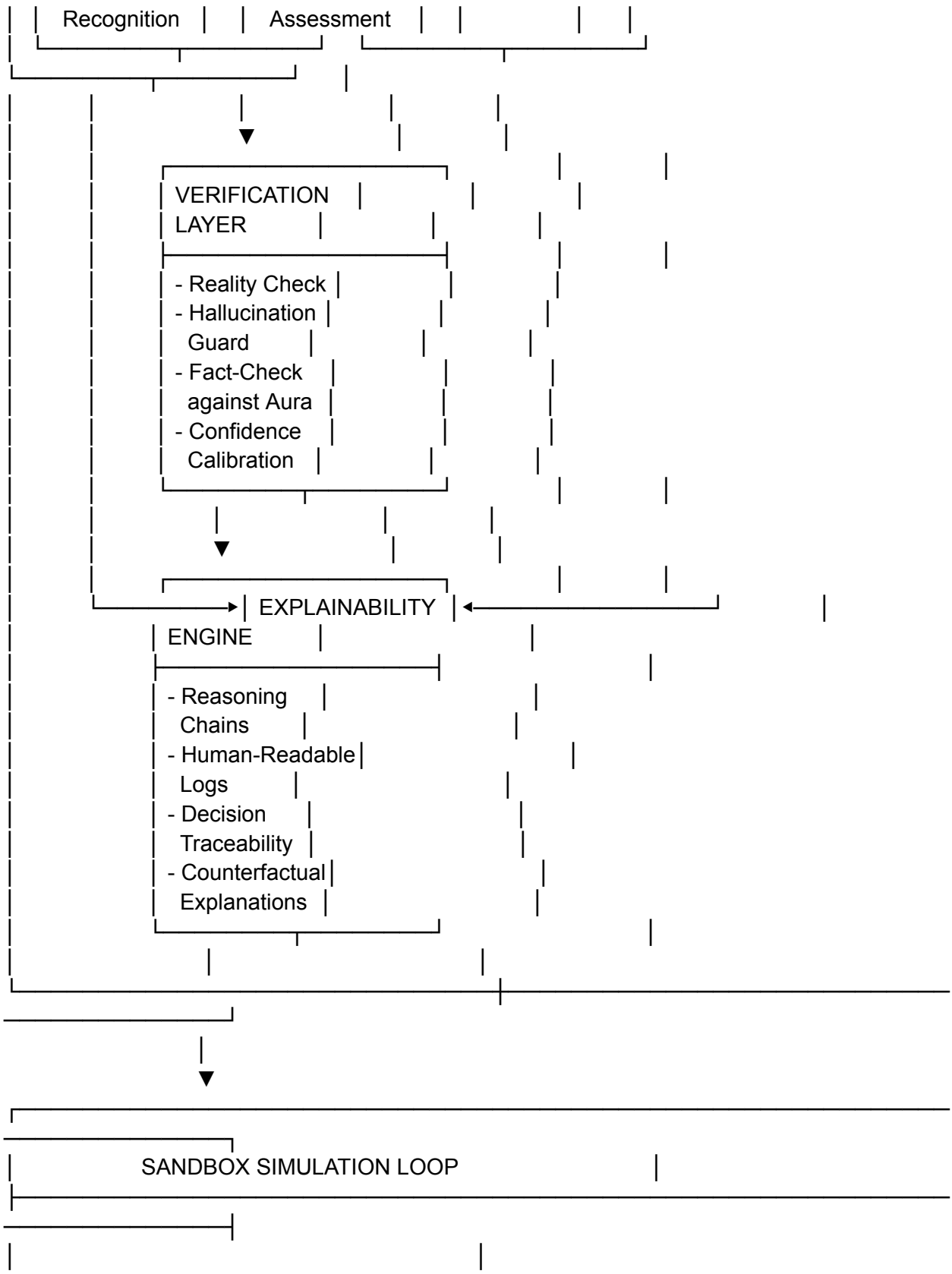
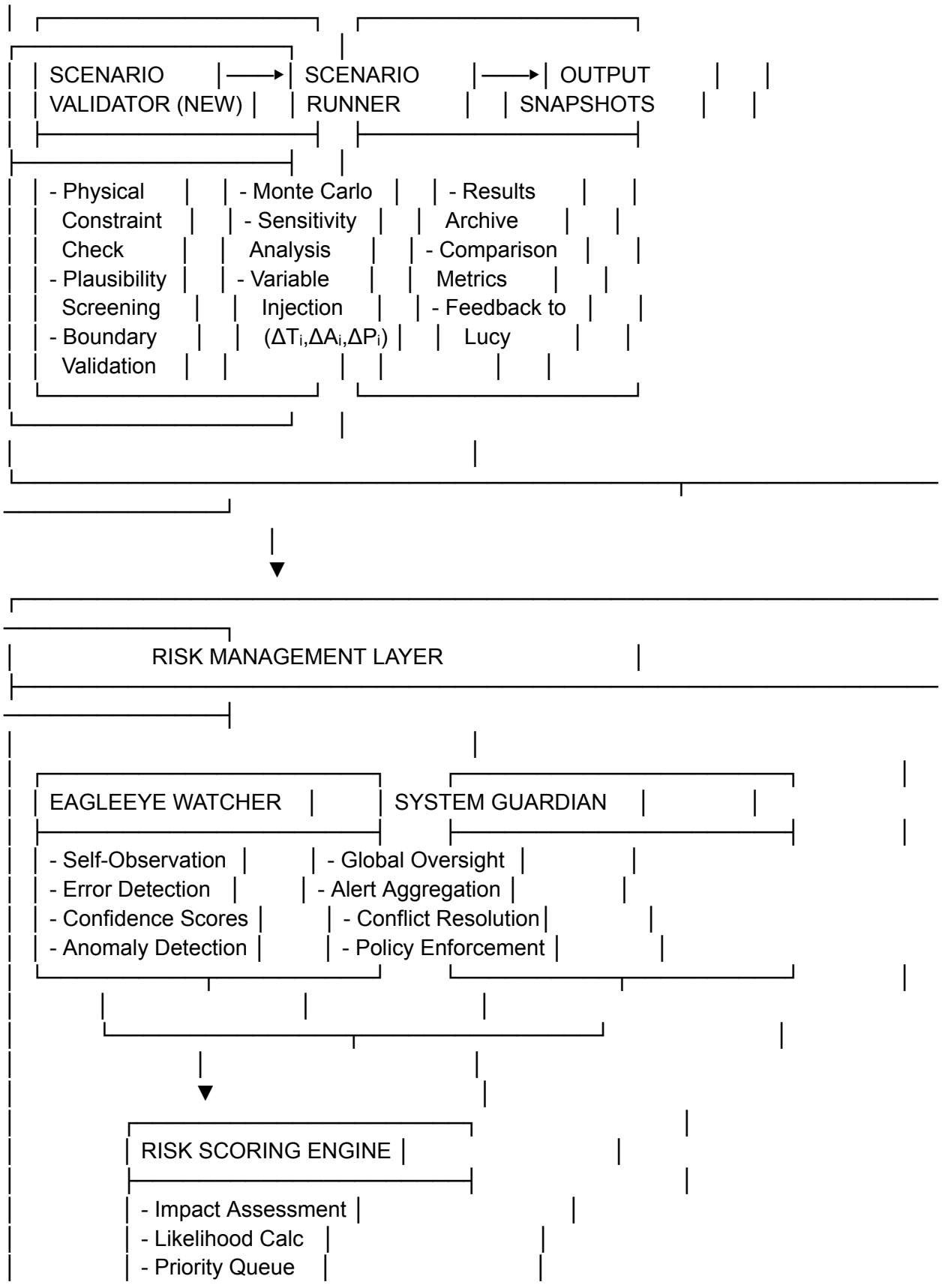
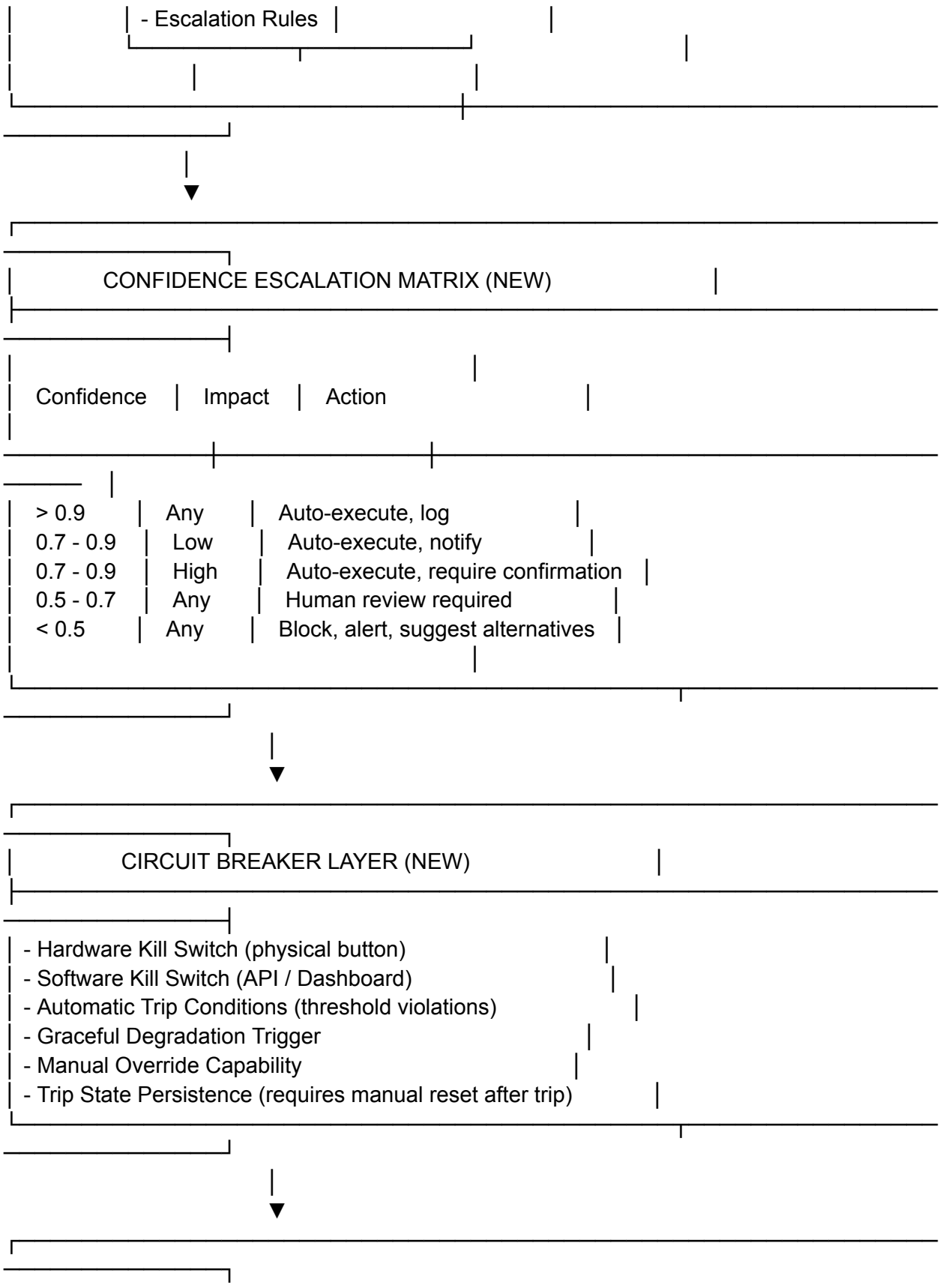
Lucy Architecture v2.0 - Enhanced

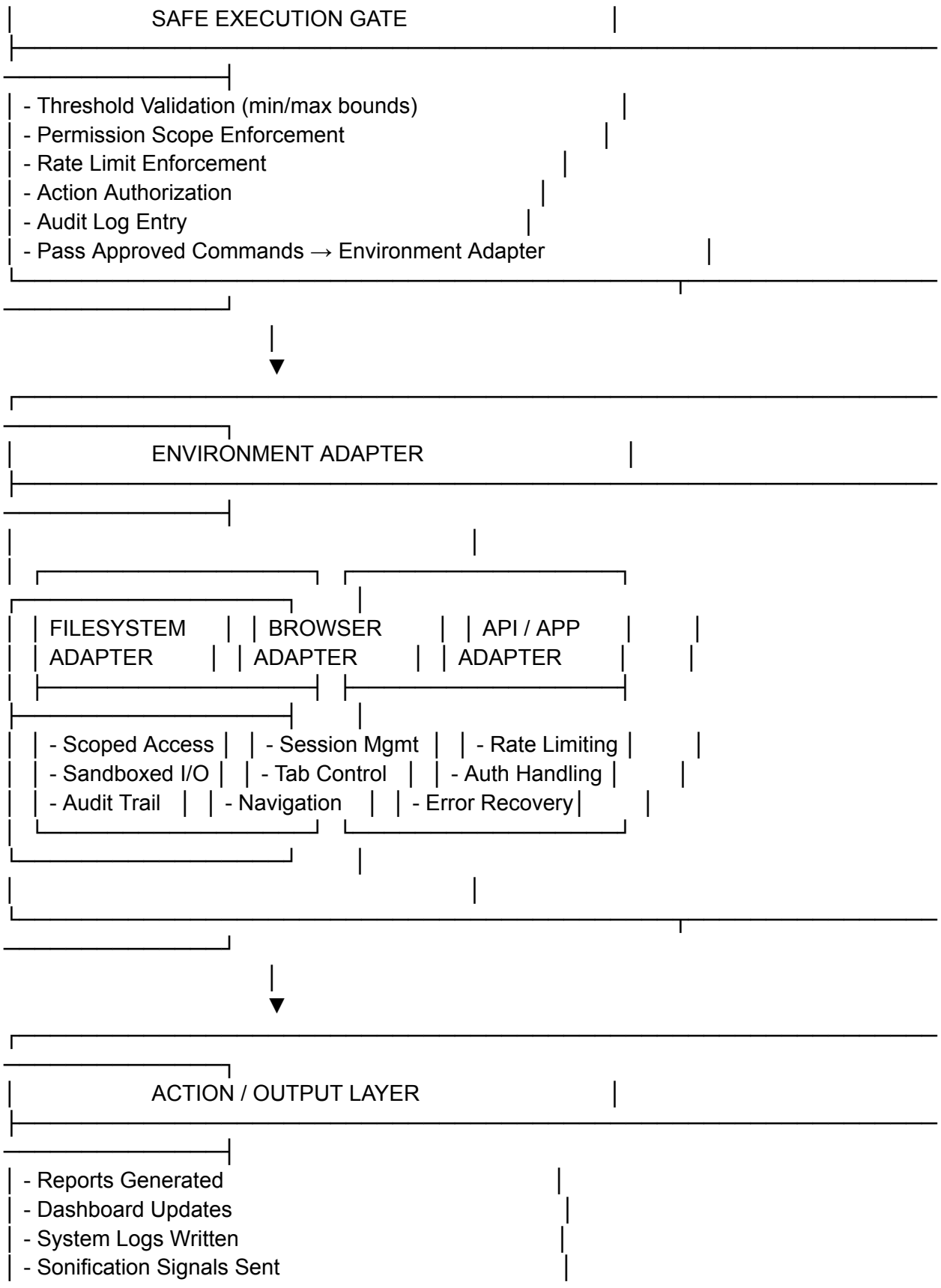


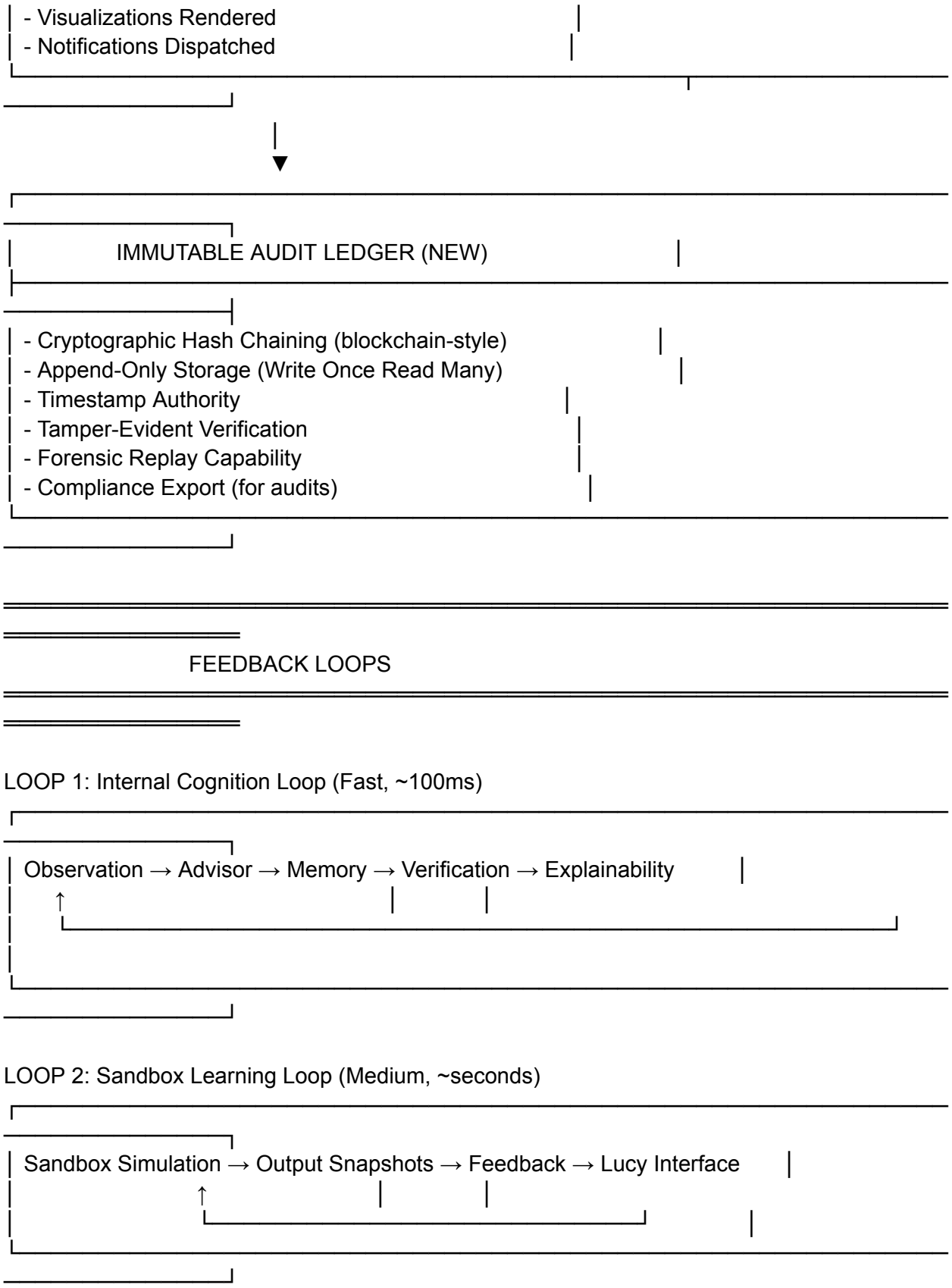

- Task Scheduling (priority queues, deadlines)
- Agent Dispatcher (spawn, coordinate, terminate)
- Pre-flight Conflict Check (resource contention)
- Agent Communication Bus (pub/sub, negotiation)
- Execution Mode Selector (auto/assisted/manual)
AURA CORE (LOCKED)
Read-Only Feed
LUCY INTERFACE
(NEW)
STATE PERSISTENCE
- KTE Equations
- Twin Earths
• Earth1 Historical
• Earth2 Accelerated
- Constants
- Thresholds
(See detailed breakdown below)
- Checkpoint every N actions
- Recovery Mode
- State Versioning
- Distributed Backup
LUCY INTERFACE (DETAILED)
OBSERVATION LAYER
ADVISOR LAYER
MEMORY ENGINE
- Read Twin Signals
- Confidence Scores
- Pattern
- Suggest Actions
- Detect Anomalies
- Risk
- Episodic
- Short-Term
- Long-Term
- Associative
Indexing


Recognition
Assessment
VERIFICATION LAYER
- Reality Check
- Hallucination Guard
- Fact-Check against Aura
- Confidence Calibration
EXPLAINABILITY ENGINE
- Reasoning Chains
- Human-Readable Logs
- Decision Traceability
- Counterfactual Explanations
SANDBOX SIMULATION LOOP

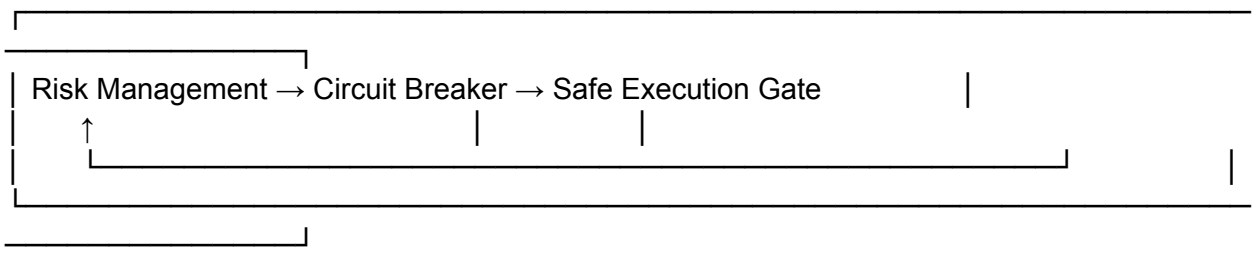

SCENARIO VALIDATOR (NEW)
SCENARIO RUNNER
OUTPUT SNAPSHOTS
- Physical Constraint Check
- Plausibility Screening
- Boundary Validation
- Monte Carlo Analysis
- Sensitivity Analysis
- Variable Injection
(ΔTi,ΔAi,ΔPi)
- Results Archive
- Comparison Metrics
- Feedback to Lucy
RISK MANAGEMENT LAYER
EAGLEEYE WATCHER
- Self-Observation
- Error Detection
- Confidence Scores
- Anomaly Detection
SYSTEM GUARDIAN
- Global Oversight
- Alert Aggregation
- Conflict Resolution
- Policy Enforcement
RISK SCORING ENGINE
- Impact Assessment
- Likelihood Calc
- Priority Queue


- Escalation Rules
CONFIDENCE ESCALATION MATRIX (NEW)
Confidence | Impact | Action
> 0.9 | Any | Auto-execute, log
0.7 - 0.9 | Low | Auto-execute, notify
0.7 - 0.9 | High | Auto-execute, require confirmation
0.5 - 0.7 | Any | Human review required
< 0.5 | Any | Block, alert, suggest alternatives
CIRCUIT BREAKER LAYER (NEW)
- Hardware Kill Switch (physical button)
- Software Kill Switch (API / Dashboard)
- Automatic Trip Conditions (threshold violations)
- Graceful Degradation Trigger
- Manual Override Capability
- Trip State Persistence (requires manual reset after trip)

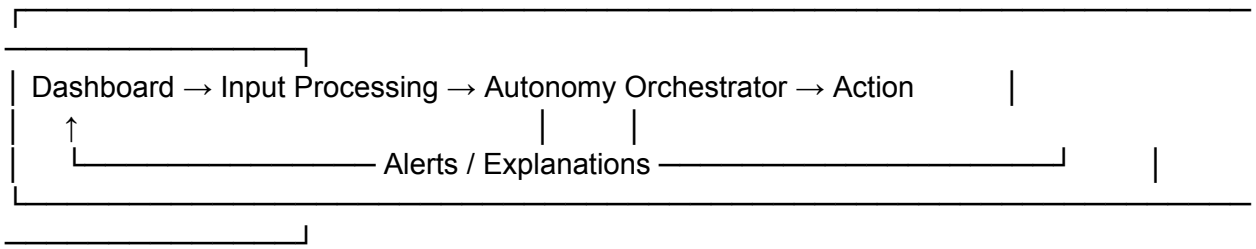

SAFE EXECUTION GATE
- Threshold Validation (min/max bounds)
- Permission Scope Enforcement
- Rate Limit Enforcement
- Action Authorization
- Audit Log Entry
- Pass Approved Commands → Environment Adapter
ENVIRONMENT ADAPTER
FILESYSTEM ADAPTER
- Scoped Access
- Sandboxed I/O
- Audit Trail
BROWSER ADAPTER
- Session Mgmt
- Tab Control
- Navigation
API / APP ADAPTER
- Rate Limiting
- Auth Handling
- Error Recovery
ACTION / OUTPUT LAYER
- Reports Generated
- Dashboard Updates
- System Logs Written
- Sonification Signals Sent


- Visualizations Rendered
- Notifications Dispatched
IMMUTABLE AUDIT LEDGER (NEW)
- Cryptographic Hash Chaining (blockchain-style)
- Append-Only Storage (Write Once Read Many)
- Timestamp Authority
- Tamper-Evident Verification
- Forensic Replay Capability
- Compliance Export (for audits)
FEEDBACK LOOPS
LOOP 1: Internal Cognition Loop (Fast, ~100ms)
Observation → Advisor → Memory → Verification → Explainability
LOOP 2: Sandbox Learning Loop (Medium, ~seconds)
Sandbox Simulation → Output Snapshots → Feedback → Lucy Interface

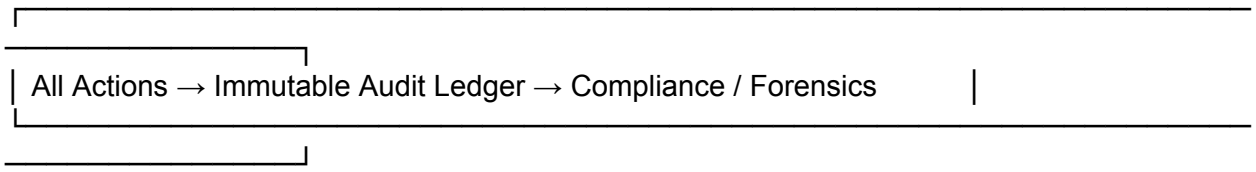
LOOP 3: Safety Enforcement Loop (Continuous)



LOOP 4: Human-in-the-Loop (On-demand)

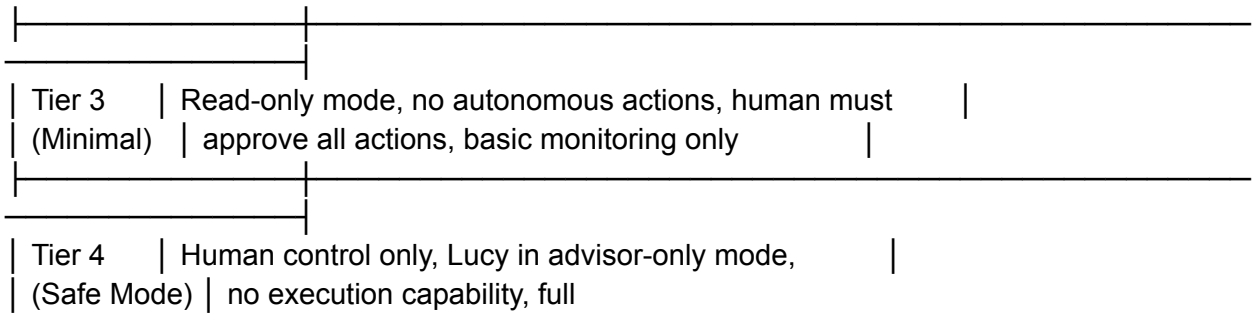


LOOP 5: Audit Trail Loop (Continuous, Asynchronous)



GRACEFUL DEGRADATION TIERS

TIER	CAPABILITIES
Tier 1 (Normal)	All features active, full autonomy, real-time simulation
Tier 2 (Degraded)	Sandbox paused, predictions cached, reduced confidence thresholds, limited concurrent tasks


Tier 3 (Minimal)
Read-only mode, no autonomous actions, human must approve all actions, basic monitoring only
Tier 4 (Safe Mode)
Human control only, Lucy in advisor-only mode, no execution capability, full